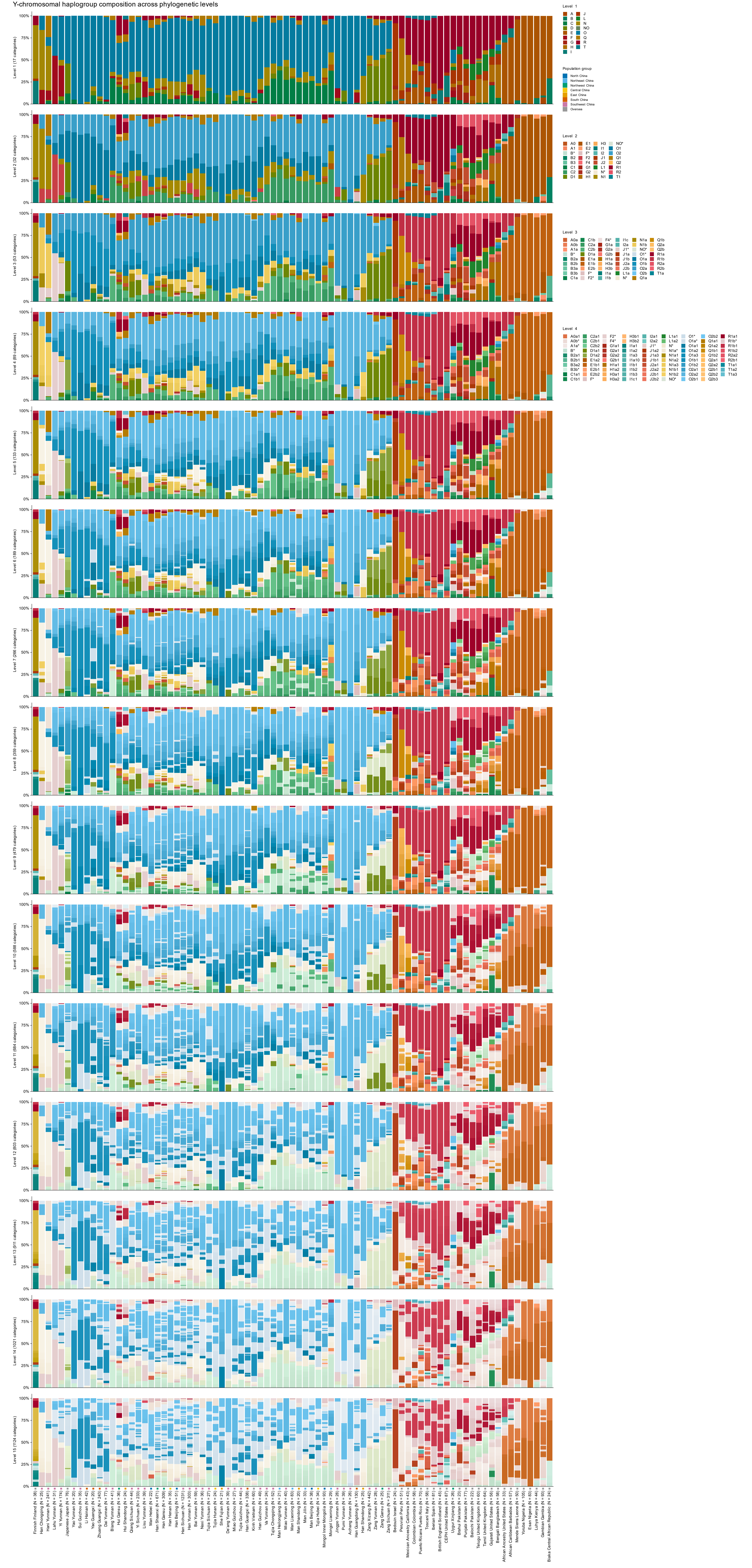
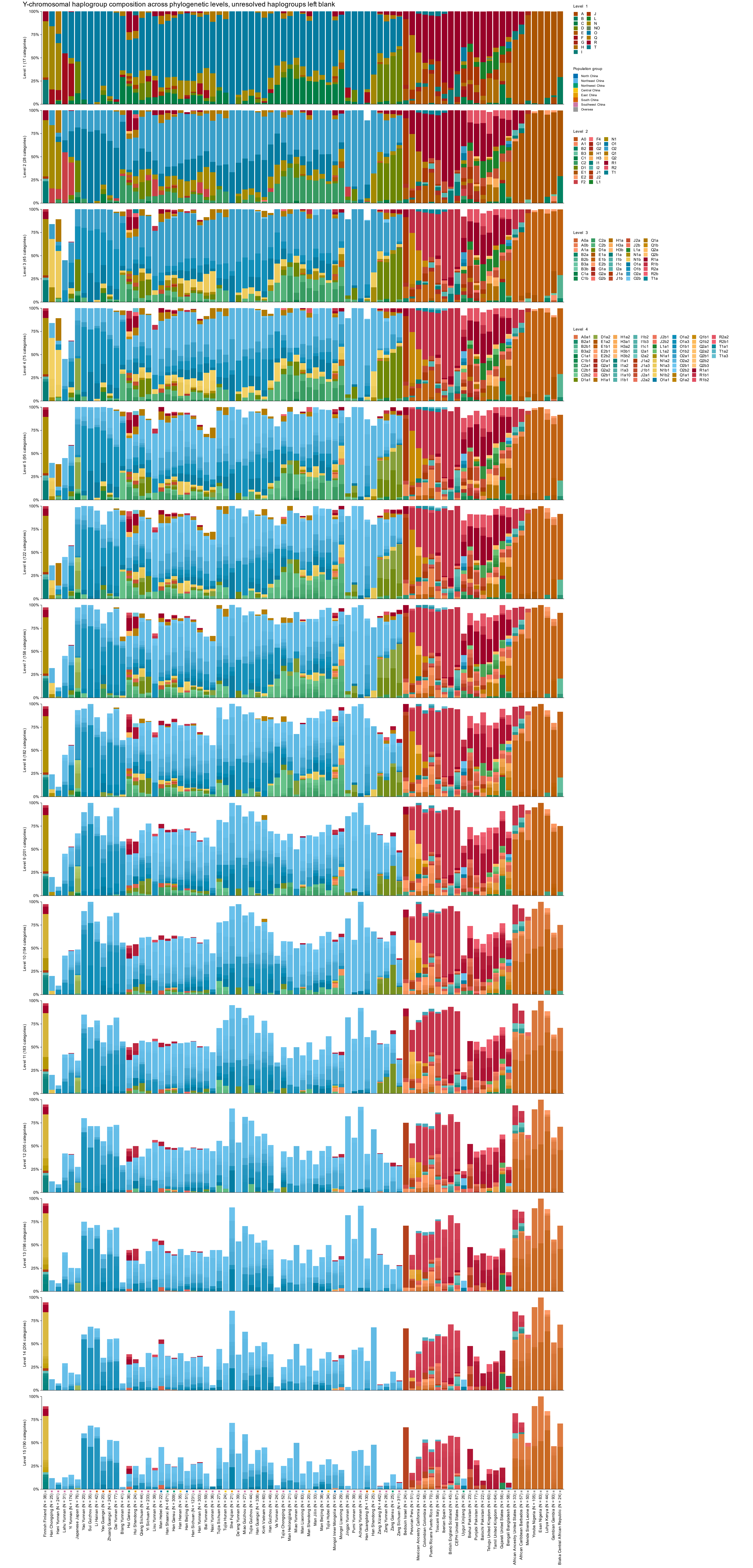


Y-chromosomal haplogroup composition across phylogenetic levels





Level 5 (133 categories)

- Aa11*
- Aa2*
- Aa3*
- B*
- Ba1a
- Ba11*
- Ba1a
- Ba2*
- Ba3*
- C1a1a
- C1a1a
- C2a11*
- C2a1a
- C2a1b
- C2a11*
- C2a1a
- C2a1b
- C2a11*
- D1a11*
- D1a1a
- D1a1b
- D1a2a
- D1a2b
- E1a2a
- E1a2b
- E1b1a
- E1b1b
- E2a1a
- E2b2a
- E2b2b
- F*
- F2*
- Fa1*
- G1a11*
- G1a1a
- G2a1a
- G2a2a
- G2a2b
- G2b1a
- H1a1a
- H1a1b
- H1a2a
- H1a2b
- H3a1a
- H3a1b
- H3a1c
- H3a2a
- H3a2b
- H3b1a
- H3b1b
- H3b2a
- Ia1b
- Ia2a
- Ia2b
- Ia3a
- Ia3b
- Ia10*
- Ib1a
- Ib2*
- Ib3*
- Ic1a
- Ia1a
- Ia1b
- Ia2b
- J1*
- J1a2a
- J1a2b
- J1a3*
- Jb1a
- Ja11*
- J2a1a
- J2a2*
- J2a2a
- J2b1b
- J2b2a
- J2b2b
- L1a11*
- L1a1a
- L1a1b
- L1a2*
- L1a2a
- N*
- N1a*
- N1a1a
- N1a2a
- N1a2b
- N1a3*
- N1b11*
- N1b1a
- N1b1b
- N1b2*
- N1b2a
- N1b2b
- N2*
- O1*
- O1a*
- O1a1a
- O1a1b
- O1a2a
- O1a3*
- O1a3a
- O1a11*
- O1b1a
- O1b2a
- O2a1a
- O2a1b
- O2a2a
- O2a2b
- O2a1a
- O2b1b
- O2a2*
- O1a1a
- O1a2a
- O1b1a
- O1b1b
- O1b2a
- O2a1a
- O2a1c
- O2a2*
- O2b1a
- O2b2a
- O2b3a
- R1a1a
- R1a*
- R1b1a
- R1b1b
- R1b2b
- R2b2b
- R2b11*
- R2b1a
- T1a1a
- T1a2b
- T1a2b

Level 6 (189 categories)

Level 8 (359 categories)

Aa11*	J2a1*
Aa5*	J2a1a1a2
Aa1a*	J2a1a1a1
B*	J2a1a1a2
Ba1a1a1	J2a1a2a2
Ba1*	J2a2*
Ba1a1a1a1	J2a2a1a1
Ba2*	J2b1a1*
Bb*	J2b2a1a*
C1a1a1a1	J2b2a1a1
C1a1a1a*	J2b2a2a*
C1a1a1a1a1	J2b2a2a1
C1a1a1a2	J2b2a2a1
C1b1a1a1	J2b2a2b2
C1b1a1a2	J2b2b1*
C1b1a1a2	L1a1*
C1b1a2b*	L1a1a1*
C2a1*	L1a1a2a*
C2a1a*	L1a1a2b*
C2a1a1a1	L1a1a1a*
C2a1a2a1	L1a1b1b*
C2a1a2a2	L1a1b2a1
C2a1a3*	L1a1b2a2
C2a1a3a*	L1a1b2b*
C2a1a3a1	L1a1b3a1
C2a1a3a2	L1a1b3a2
C2a1a3a6	L1a1b3b*
C2a1a4a*	L1a1b3b1
C2a1b*	L1a2*
C2a1*	L1a2a1a*
C2b1a1*	L1a2a1a1
C2b1a1a1	L1a2a1b*
C2b1a1a2	L1a2a1b1
C2b1a2a1	L1a2a1b2
C2b1a2a2	L1a2a2*
C2b1a2b*	N*
C2b1a2b1	N1a*
C2b1a2b2	N1a1a1a1
C2b1b1*	N1a1a1a*
C2b1b1a1	N1a1a1a*
C2b1b1a2	N1a2a1*
C2b1b1b1	N1a2a1a1
C2b1b1b2	N1a2a1a2
C2b1b1b3	N1a2a1b*
C2b1b1b4	N1a2a1b1
C2b1b1b5	N1a2b3*
C2b1b1b*	N1a3*
C2b1b1a*	N1a31*
C2b1b1a1	N1b1a1*
C2b1b1a*	N1b1a1a*
C2b1b1b1	N1b1a1b*
C2b1b1b2	N1b1a2*
C2b1b1b*	N1b1a2a*
C2b1b1*	N1b1a2b*
C2b1b2*	N1b1a2b1
C2b2*	N1b1b*
D1a1*	N1b1b1a1
D1a1a1a1	N1b2*
D1a1a1a1	N1b2a*
D1a1a1a2	N1b2a1a*
D1a1b1a*	N1b2a1b*
D1a1b1a1	N1b2b*
D1a1b1a3	Nb*
D1a2a1b*	Or*
D1a2a1a2	Or1a*
D1a2a1a1	Or1a1a1a1
D1a2a2a1	Or1a1a1a2
D1a2a2a2	Or1a1a1b*
D1a2a2a3	Or1a1a1b1
D1a2a2b*	Or1a1a2a1
D1a2b1a2	Or1a1a2a2
D1a2b1a1	Or1a1a2a3
D1a2b1b2	Or1a1a2b1
D1a2b2*	Or1a1a2b2
E1a2a1a1	Or1a1a2b*
E1a2a1a2	Or1a1b*
E1a2a1a1	Or1a1b1
E1a2a1a2	Or1a1b1a1
E1a2a1*	Or1a2a*
E1a2a2*	Or1a3*
E1a2b*	Or1a3a*
E1a2b1*	Or1b1*
E1a2b1a1	Or1b1a*
E1a2b1a2	Or1b1a1a1
E1a2b1b*	Or1b1a1a2
E1b1a1a1	Or1b1a1b*
E1b1a1a1	Or1b1a1b1
E1b1a1a2	Or1b1a2a*
E1b1b1a1	Or1b1a2a1
E1b1b1a2	Or1b1a2b1
E1b1b1b1	Or1b1a2b2
E1b1b1b2	Or1b1a2b*
E2a1a1a*	Or1b2a1a*
E2b1a1a*	Or1b2a1a1
E2b1a2*	Or1b2a1a2
E2b1a*	Or1b2a1a3
E2b2*	Or1b2a1b*
F*	Q2a1a*
F2*	Q2a1a1a1
G1a1*	Q2a1a1a2
G1a1a2*	Q2a1a1b1
G1a1a2a1	Q2a1a1b2
G1a1a2b*	Q2a1a1a1
G2a1a2a*	Q2a1a1a2
G2a1a2a1	Q2a1b1*
G2a2a1a2	Q2a1b1b1
G2a2a1a3	Q2a1b1b*
G2a2a1b1	Q2a1b2a1
G2a2a1a1	Q2a1b2a2
G2a2b2a1	Q2a2a1a1
G2a2b2a4	Q2a2a1a2
G2a2b2b2	Q2a2a1b*
G2a1a*	Q2a2a2*
H1a1a1a*	Q2a2a2a1
H1a1a1a*	Q2a2a2a2
H1a1a4a*	Q2a2b1*
H1a1a4a1	Q2a2b1a1
H1a1a4a2	Q2a2b1a2
H1a1a4a*	Q2a2b2a1
H1a1a4b1	Q2a2b2a2
H1a1a4b2	Q2a2b2b*
H1a1a4b3	Q2a2b2b1
H1a1a4b4	Q2b1a*
H1a1a4b6	Q2b1a1*
H1a1a4b7	Q2b1a1a*
H1a1b1a*	Q2b1a1a1
H1a1b1a1	Q2b1b*
H1a1b1b1	Q2b2*
H1a1b2*	Q1a1a*
H1a2a2*	Q1a1a1*
H1a2b*	Q1a1a1a*
H1a2b*	Q1a1a1a1
H1a2b1a1	Q1a1a1b1
H1a2b1a2	Q1a1a2*
H1a2b1b1	Q1a2a2a1
H1a2b1b2	Q1a2a2b*
H3a1a*	Q1a2a2b1
H3a1b*	Q1b1a1a1
H3a1c*	Q1b1a2a1
H3a2a1a1	Q1b1a3a*
H3a2a1a*	Q1b1a3a1
H3a2a2a*	Q1b1a3b*
H3a2b*	Q1b1a3b2
H3a2b1*	Q1b1a3b3
H3a2b2*	Q1b1b1*
H3b1a*	Q1b2a1a1
H3b1b*	Q1b2a2*
H3b2a*	Q2a1a4*
Ia1b1a2	Q2a1c1b1
Ia1b1a4	Q2a2*
Ia1b1b1	Q2b1a1*
Ia2a1a1	Q2b2a*
Ia2a2a2	Q2b2a1a*
Ia2a2a3	Q2b2a*
Ia2b1a*	R1a1a1a1
Ia2b2*	R1a1a1a1
Ia3a1a*	R1a1a1b2
Ia3b2a2	R1b*
Ia3b3a*	R1b1a1a1
Ia1b*	R1b1a1b1
Ia1a1*	R1b1a2*
Ib2*	R1b1a2a1
Ib3*	R1b1a2b1
Ic1a1a1	R1b1b2a2
Qa1a1a1	R1b2b*
Qa1a1b1	R1b2b1*
Qa1a2b1	R1b2b2*
Qa1b1a1	R2a2b1*
Qa1b1a2	R2a2b1b1
Qa1b2a2	R2a2b1b2
Qa2b1a2	R2b1*
J1*	R2b1a1*
J1a2a1a2	T1a1a*
J1a2a1b1	T1a1a1b2
J1a2b1b2	T1a2b1a*
J1a2*	T1a2b1a1
J1b1a1a*	T1a2b2a1
J1b1a1b*	

Level 10 (588 categories)

Aa11*	H1a1a42b2	O1b1*
Ab2*	H1a1a42c1	O1b1a*
A1a*	H1a1a42c3	O1b1a1a1
B*	H1a1a42c3	O1b1a1a1a1
B2a1a1a1a*	H1a1a42c*	O1b1a1a1a2
B2a1a1a1a1	H1a1a42c1	O1b1a1a1a2*
B2a1a1a1a2	H1a1a43a2	O1b1a1a1b1
B2a1*	H1a1a43b*	O1b1a1a1a2
B2b1a1c1*	H1a1a43b1	O1b1a1a2*
Ba2*	H1a1a44a*	O1b1a1b*
Bb2*	H1a1a44b*	O1b1a1b1a*
C1a1a1a1a*	H1a1a44b*	O1b1a1b1a1
C1a1a1a1b*	H1a1a44b*	O1b1a1b1a*
C1a1a1b*	H1a1b1a*	O1b1a1b1b1
C1a1a1b1*	H1a1b1a1a*	O1b1a2a*
C1a1a1b2*	H1a1b1a1b*	O1b1a2a1*
C1b1a1a1a1	H1a1b1b1*	O1b1a2a1a*
C1b1a1b*	H1a1b2*	O1b1a2b1*
C1b1a1a2a1	H1a2a2*	O1b1a2b1a*
C1b1a1a2a2	H1a2a3*	O1b1a2b2*
C1b1a1a2b*	H1a2b*	O1b1a2c*
C1b1a2b*	H1a2b1a1*	O1b2a1a*
C2a1*	H1a2b1a2*	O1b2a1a1*
C2a1a*	H1a2b1b1*	O1b2a1a1a*
C2a1a1b1a*	H3a1a*	O1b2a1a1c*
C2a1a1b1a1	H3a1b*	O1b2a1a2a*
C2a1a1b1a2	H3a1c1*	O1b2a1a2a1
C2a1a1b1a3	H3a2a1a1a*	O1b2a1a2b*
C2a1a1a1a4	H3a2a1a*	O1b2a1a2b*
C2a1a1b1b*	H3a2a2a*	O1b2a1b*
C2a1a2a1a1	H3a2a2b*	O2a1a*
C2a1a2a2a2	H3a2b1*	O2a1a1a1a*
C2a1a2a2b*	H3a2b2*	O2a1a1a1a1
C2a1a3*	H3b1a*	O2a1a1a1a2
C2a1a3a*	H3b1b*	O2a1a1a1b1
C2a1a3a1*	H3b2a*	O2a1a1a1b2
C2a1a3a1a1	H41b1a2a1	O2a1a1a2*
C2a1a3a2a*	H41b1a4a1	O2a1a1a2a1
C2a1a3a2a1	H41b1a4a2	O2a1a1a2a2
C2a1a3a3*	H41b1b1a1	O2a1a1b1a*
C2a1a3a3a*	H4a2a1a1	O2a1a1b1a1
C2a1a4a*	H4a2a1a2	O2a1a1b1a2
C2a1b*	H4a2a1a4	O2a1a1b1a3
C2a1b1*	H4a2a1a5	O2a1a1b1b*
C2b1a1*	H4a2a1a1a1	O2a1a1b2*
C2b1a1a1*	H4a2a1a1a2	O2a1a1b1a1
C2b1a1a1a2	H4a2a2a*	O2a1a1b1a*
C2b1a1a1b*	H4a2a2a1a1	O2a1a1b2a1
C2b1a1a2*	H4a2b1a*	O2a1b1a2b*
C2b1a2a1*	H4a2b*	O2a1b1a2b1
C2b1a2a1a*	H4a3a1a*	O2a1b1a2b2
C2b1a2a1a1	H4b2a2a*	O2a1b1b*
C2b1a2a1a2	H4b3a*	O2a1b1b1*
C2b1a2a1b*	H41b1*	O2a1b1b2*
C2b1a2a2*	Hb2*	O2a1b2a1*
C2b1a2a2b1	Hb3*	O2a1a2a1a*
C2b1a2a2b2	Hc1a1a1a*	O2a1b2a1a*
C2b1a2b*	Oa1a1a1*	O2a1b2a2*
C2b1a2b1*	Oa1a1a1a1	O2a1b2a2a1
C2b1a2b2*	Oa1a1a1a2	O2a2a1a1a*
C2b1a2b2c*	Oa1a1a1a2	O2a2a1a1a1
C2b1a2b2c*	Oa1a1a1a1	O2a2a1a1a*
C2b1a2b1*	Oa1b1a1a1	O2a2a1a1b1
C2b1a2b1b*	Oa1b1a1a1	O2a2a1a1a2
C2b1b1b*	Oa1b1a2b1	O2a2a1a2a1
C2b1b1b1*	Oa1b2a2c2	O2a2a1a2a3
C2b1b1b2*	Oa1b2c1c2	O2a2a1b*
C2b1b1b2a1	J1*	O2a2a2*
C2b1b1b2a2	J1a2a1a2b1	O2a2a2a1*
C2b1b1b3*	J1a2a1a2b2	O2a2a2a1a1
C2b1b1b4*	J1a2b1b1a1	O2a2a2a2*
C2b1b1b5*	J1a2b1b2*	O2a2a1*
C2b1b1c*	J1a2b1b2a*	O2a2b1a1a*
C2b1b1c*	J1a2b1b2b*	O2a2b1a1a*
C2b1b1d*	J1a2b1b2c*	O2a2b1a1a1
C2b1b1d1*	J1a3*	O2a2b1a1b*
C2b1b1d1a*	J1b1a1a*	O2a2b1a2*
C2b1b1d1*	J1b1a1b*	O2a2b1a2a*
C2b1b1d1b*	J2a1*	O2a2b1a2a1
C2b1b1d1c*	J2a1a1a1a1	O2a2b1a2a2
C2b1b1*	J2a1a1a2a2	O2a2b2a1*
C2b1b1*	J2a1a1a2b1	O2a2b2a1a*
C2b1b2*	J2a1a1a2b2	O2a2b2a1a1
C2b1*	J2a1a1b1*	O2a2b2a1a2
D1a1*	J2a1a1b1a*	O2a2b2a1a*
D1a1a1a1a1*	J2a1a1b1a1	O2a2b2a1b1
D1a1a1a1a1*	J2a1a1b1b*	O2a2b2a1a2
D1a1a1a1b2*	J2a1a1b2a1	O2a2b2a2a*
D1a1a1b1b*	J2a1a2b2a2	O2a2b2a2a1
D1a1a1b2*	J2a2*	O2a2b2a2b*
D1a1b1a*	J2a2a1a1a*	O2a2b2a2b1
D1a1b1a1*	J2b1b1*	O2a2b2a2b1
D1a1b1a3*	J2b2a1a*	O2a2b2b1*
D1a2a1*	J2b2a1a1a1	O2a2b2b1a*
D1a2a1b2*	J2b2a2a*	O2a2b2b1a1
D1a2a1b2a1	J2b2a2a1*	O2b1a*
D1a2a1b2a2	J2b2a2b1a1	O2b1a1*
D1a2a1c1a1*	J2b2a2b2*	O2b1a1a*
D1a2a1c1b*	J2b2a2b2a*	O2b1a1a1
D1a2a1d1a1*	J2b2a2b2b*	O2b1a1a1a1
D1a2a2a1a1*	J2b2a2b2b1	O2b1b*
D1a2a2a1a2	J2b2b1*	O2b2*
D1a2a2a2a*	L1a1*	O1a1a*
D1a2a2a3a2	L1a1a1*	O1a1a1*
D1a2a2a3a2	L1a1a2a*	O1a1a1a*
D1a2a2b*	L1a1a2b*	O1a1a1a1*
D1a2b1a1a1	L1a1b1a*	O1a1a1a1a1
D1a2b1a2*	L1a1b1b*	O1a1a1b1*
D1a2b2*	L1a1b2a2*	O1a1a2*
El2a1a1a1	L1a1b2b*	O1a1a2a1*
El2a1a1a2	L1a1b3a1*	O1a1a2b1*
El2a1a1a2b*	L1a1b3a1a2	O1a1a1a1a2
El2a1a1a2b*	L1a1b3a1b*	O1a1a1a1a2
El2a1a1a*	L1a1b3a2a*	O1a1a1a1a*
El2a1a1b*	L1a1b3a2b*	O1a1a1a1a*
El2a1a1c*	L1a1b3a2c*	O1a1a1a1b1
El2a1a1c*	L1a1b3b*	O1a1a1a1a2
El2a2a*	L1a1b3c1*	O1a1a1a1b1
El2a2b*	L1a2*	O1a1a1a1b*
El2a2b*	L1a2a1a*	O1a1a1a1b1
El2a2b1*	L1a2a1a1a2	O1a1a1a1b1
El2a2b1a1a1	L1a2a1a1b*	O1a1a1a1b1
El2a2b1a1a2	L1a2a1b*	O1a1a1a1b1
El2a2b1a2*	L1a2a1b1*	O1a1a1a1b1
El2a2b1a2b*	L1a2a1b2*	O1a1a1a1b1
El2a2b1c*	L1a2a2*	O1a1a1a1b1
Elb1a1a1a1	N*	O1a1a1a1c*
Elb1a1a1a2	N1a*	O1a1a1a1a*
Elb1a1a1a3	N1a1a1a1a1	O1a1a1a1a*
Elb1a1a1b*	N1a1a1a1a2	O1a1a1a1a*
Elb1a1a1a1	N1a1a1a1a3	O1a1a1a1c*
Elb1a1b1a2	N1a1a1a1a4	O1a1a2a1b1
Elb1a1b1a3	N1a1a2*	O1a1a2a1b2
Elb1a1b1b1	N1a2a*	O1b1a2a*
Elb1a1b1b2	N1a2a1*	O1b1a2a1*
Elb1a1b1c1	N1a2a1a1*	O1b1a2b*
Elb1a1b1c2	N1a2a1a2*	O1b1a2b2*
Elb1a1b1d*	N1a2b1*	O1b1a2b2*
Elb1a1b1e*	N1a2b1b1*	O1b1b1*
Elb1a1b1f	N1a2b3*	O1b2a1a*
Elb1a1b1g*	N1a3*	O1b2a1a1a*
Elb1b1a1*	N1b1*	O1b2a2*
Elb1b1a1a1	N1b1a1*	O2a1a4*
Elb1b1a1a2	N1b1a1a*	O2a1c1b1*
Elb1b1a1b1	N1b1a1b*	O2a2*
Elb1b1a1b2	N1b1a2*	O2b1a1*
Elb1b1a2a*	N1b1a2a*	O2b2a*
Elb1b1a1a1	N1b1a2b*	O2b2a1a*
Elb1b1b1b*	N1b1a2b1*	O2b2a1a*
Elb1b1b1b1	N1b1a2b1a*	R1a1a1a1b1
Elb1b1b1b2	N1b1b*	R1a1a1b1a1
Elb1b1b2a1	N1b1b1*	R1a1a1b1a2
Elb1b1b2b2	N1b2*	R1a1a1b1a3
Elb1a1c*	N1b2a*	R1a1a1b1a4
E2b1a1a*	N1b2a1a*	R1a1a1b2*
E2b1a2*	N1b2a1b*	R1a1a1b2a1
E2b2*	N1b2b*	R1a1a1b2a2
F*	Nb*	R1a1a1b2a3
F1*	O1*	R1a1a1b2a4
F2*	O1a*	R1a1a1b2a1
F4*	O1a1a1a1*	R1a1a1b2b*
G1a1*	O1a1a1a1a*	R1b*
G1a1a2*	O1a1a1a1a1	R1b1a1a1a*
G1a1a2a1*	O1a1a1a1a2	R1b1a1a1b*
G1a1a2b*	O1a1a1a2*	R1b1a1a1b1
G2a1a2a*	O1a1a1a3a*	R1b1a1b1a1
G2a1a2a1a*	O1a1a1b*	R1b1a1b1a2
G2a2a1a2a1	O1a1a1b1*	R1b1a1b1a*
G2a2a1a3*	O1a1a1b1a1	R1b1a1b1b3
G2a2b1a1*	O1a1a1b1b*	R1b1a2*
G2a2b1a1a1	O1a1a1b1b1	R1b1a2a1a*
G2a2b1b1b1	O1a1a2a1*	R1b1a2a1c*
G2a2b2b1*	O1a1a2a1a*	R1b1a2b1*
G2a2b2a1a1	O1a1a2a1a1	R1b1b2a2b1
G2a2b2a1a*	O1a1a2a1a2	R1b2b*
G2a2b2a1a2	O1a1a2a1a3	R1b2b1*
G2a2b2b2a*	O1a1a2a1b1	R1b2b2*
G2a2b2b2a*	O1a1a2a2*	R2a2b1*
H1a1a1b*	O1a1a2a2a1	R2a2b1b1*
H1a1a1c*	O1a1a2a2*	R2a2b1b2a1
H1a1a1a*	O1a1a2b1*	R2a2b1b2b2
H1a1a1a1*	O1a1a2b2*	R2a2b1b2b3
H1a1a1a2*	O1a1a2b2a*	R2a2b1b2b*
H1a1a1a3*	O1a1a2c*	R2a2b1b2c1
H1a1a1a1a*	O1a1b*	R2b1*
H1a1a4b1b*	O1a1b1a1a*	R2b1a1*
H1a1a4b2*	O1a1b1a2*	T1a1a*
H1a1a4b2a*	O1a2a*	T1a1a1b2b2
H1a1a4b2a1	O1a2a2*	T1a2b1a*
H1a1a4b2a2	O1a3*	T1a2b1a1*
H1a1a4b2b1	O1a3a*	T1a3b2a1*

Level 11 (684 categories)

Aa11*	H1a1a4b2b*	O1b1a1b1b*
Aa2*	H1a1a4b3b1a	O1b1a1b1b1a
Aa3*	H1a1a4b3b1b	O1b1a1b1b1b
B*	H1a1a4b3b1c	O1b1a1b2a1*
Ba1a1a1a*	H1a1a4b4a*	O1b1a1b2a1*
Ba1a1a1a1*	H1a1a4b4b*	O1b1a1b2b1*
Ba1a1a1a2*	H1a1a4b4b*	O1b1a1b2b1a*
Ba1a1a1a2a*	H1a1a4b7*	O1b1a1b2b1*
Ba1a1a1a2a	H1a1b1a*	O1b1a1b2c*
Bb11*	H1a1b1a*	O1b1a1b2c*
Bb1a1a11*	H1a1b1a1a*	O1b2a1a*
Bb2*	H1a1b1a1b*	O1b2a1a1*
Bb3*	H1a1b1b1*	O1b2a1a1a*
C1a1a1a1a*	H1a1b2*	O1b2a1a1a1*
C1a1a1a1b*	H1a2a2*	O1b2a1a1c*
C1a1a1a1b	H1a2a3*	O1b2a1a2a*
C1a1a1b1*	H1a2b*	O1b2a1a2a1*
C1a1a1b2*	H1a2b1a1*	O1b2a1a2a1a
C1a1a1a1a1a	H1a2b1a2*	O1b2a1a2a1b
C1b1a1a1a1b	H1a2b1b1*	O1b2a1a2b*
C1a1a1a1b1*	H1a2b1b2*	O1b2a1a2b*
C1b1a1a2a1a	H3a1a*	O1b2a1a*
C1b1a1a2a1b	H3a1b*	O2a1a*
C1b1a1a2b*	H3a1c*	O2a1a1a1a*
C1b1a1a2b*	H3a2a1a1a*	O2a1a1a1a1a
C1b1a2*	H3a2a1a*	O2a1a1a1a1b
C1b1a2b*	H3a2a2a*	O2a1a1a1a2*
C2a1*	H3a2a2b*	O2a1a1a1b1*
C2a1a*	H3a2b1*	O2a1a1a1b1a
C2a1a1b1a*	H3a2b2*	O2a1a1a1b2*
C2a1a1a1a1*	H3a1a*	O2a1a1a1b2a
C2a1a1b1a2a	H3b1b*	O2a1a1a2*
C2a1a1b1a3a	H3b2a*	O2a1a1a2a1*
C2a1a1a1a4*	H4b1b1a2a1*	O2a1a1a2b2*
C2a1a1b1b1*	H4b1b1a4a1a	O2a1a1b1a*
C2a1a1a1a1*	H4b1b1a4b1a	O2a1a1b1a1*
C2a1a2a2a2*	H4b1b1a4a2b	O2a1a1b1a1a
C2a1a2a2b*	H4b1b1a1a1a	O2a1a1a1b1b
C2a1a2*	H4a1a1a1a	O2a1a1a1b2a
C2a1a3a*	H4a2a1a1a1b	O2a1a1b1a2b
C2a1a3a1*	H4a2a1a1a2a	O2a1a1b1a2*
C2a1a3a1a1*	H4a2a1a1a2b	O2a1a1b1b*
C2a1a3a2*	H4a2a1a1a4a	O2a1a1b2*
C2a1a3a2a1*	H4a2a1a1a4a	O2a1a1b1a1a
C2a1a3a3*	H4a2a1a1a1a	O2a1a1b1a1b
C2a1a3a3b*	H4a2a1a1a1a	O2a1a1b1a1b
C2a1a3a4*	H4a2a1a1a2a	O2a1a1b1*
C2a1a4b*	H4a2a2a2*	O2a1a1a2a1a
C2a1b*	H4a2a2a1*	O2a1a1a2a2a
C2a1*	H4a2a2a2a2*	O2a1a1a2a2b
C2b1a1*	H4a2b1a*	O2a1a1a2b*
C2b1a1a1*	H4a2b2*	O2a1a1a2b1*
C2b1a1a1a1*	H4a3a1a*	O2a1a1a2b2a
C2b1a1a1a2*	H4b3a*	O2a1a1a2b2b
C2b1a1a1a3*	H4b3b*	O2a1b1b*
C2b1a1a1b*	H4b3*	O2a1b1b1*
C2b1a1a2*	H4b4*	O2a1b1b2*
C2b1a2a1*	H5a*	O2a1b2*
C2b1a2a1a1*	H5a1a1a*	O2a1b2a1a*
C2b1a2a1a2*	H4a1a1a1a	O2a1b2b1*
C2b1a2a1b*	H4a1a1a1a	O2a1b2b2*
C2b1a2a2*	H4a1a1a1a2b	O2a1b2b2a*
C2b1a2a2b*	H4a1a1a1a1a	O2a1b2b2a1*
C2b1a2a2b1*	H4a1a1a1a1a	O2a1b2b2a1*
C2b1a2a2b1a	H4b1a1a1a	O2a1b2b2a1a
C2b1a2a2b1b	H4b1a1a1a	O2a1b2b2a1b
C2b1a2a2b1c	H4b1a2b1a	O2a1b2b2a1c
C2b1a2a2b2*	H4b1a2b2a	O2a1b2b2a1*
C2b1a2b*	O2b1b1c2*	O2a2a1a1b1a
C2b1a2b1*	J1*	O2a2a1a1b2*
C2b1a2b2*	J1a2a1a2c1a	O2a2a1a1b2b
C2b1a2b2a*	J1a2a1a2b2b	O2a2a1a2a1a
C2b1a2b2b*	J1a2a1a2b1a1*	O2a2a1a2a1b
C2b1b1*	J1a2b1b2*	O2a2a1a2a2*
C2b1b1a1b*	J1a2b1b2a*	O2a2a1a2a2a
C2b1b1b*	J1a2b1b2c*	O2a2a1*
C2b1b1b1*	J1a2*	O2a2a2*
C2b1b1b2*	J1a1a1a*	O2a2a2a1*
C2b1b1b2a1*	J1b1a1b*	O2a2a2a1a1*
C2b1b1b2a2*	J2a1*	O2a2a2a2a1a
C2b1b1b2a2*	J2a1a1a2a1a	O2a2a2a2*
C2b1b1b3*	J2a1a1a2a2b	O2a2b1*
C2b1b1b4*	J2a1a1a2b1b	O2a2b1a1a*
C2b1b1b5*	J2a1a1a2b2a	O2a2b1a1a1*
C2b1b1a*	J2a1a1b1a*	O2a2b1a1a1b
C2b1b1a1*	J2a1a1b1a1a	O2a2b1a1a1c
C2b1b1a2*	J2a1a1b1a1*	O2a2b1a1a1d
C2b1b1a3*	J2a1a1b2a1*	O2a2b1a1a1e
C2b1b1a4*	J2a1a1b2a1a	O2a2b1a1a1g
C2b1b1a5*	J2a1a1b2a1b	O2a2b1a1a1h
C2b1b1*	J2a1a1b2a2*	O2a2b1a1b*
C2b1b1*	J2a1a1b2a2b	O2a2b1a1c*
C2b1b2*	J2a2*	O2a2b1a2a*
C2a2*	J2a2a1a1a*	O2a2b1a2a1*
D1a1*	J2b1b1*	O2a2b1a2a1a
D1a1a1a1a1*	J2b2a1*	O2a2b1a2a2*
D1a1a1a1b*	J2b2a1a1a*	O2a2b1a2a2a
D1a1a1a1b2*	J2b2a2*	O2a2b1a2a2b
D1a1a1a1b*	J2b2a2a1*	O2a2b1a1*
D1a1a1a2*	J2b2a2b1a1c	O2a2b2a1a*
D1a1b1*	J2b2a2b2*	O2a2b2a1a1*
D1a1b1a1*	J2b2a2b2a*	O2a2b2a1a2*
D1a1b1a3*	J2b2a2b2b*	O2a2b2a1a*
D1a1b1a*	J2b2a2b2b1a	O2a2b2a1b1*
D1a2a1a2*	J2b2a2b2b1b	O2a2b2a1b2*
D1a2a1a2a1a	J2b2b1*	O2a2b2a1b2a
D1a2a1a2a1b	L1a1*	O2a2b2a2a*
D1a2a1a2a2*	L1a1a1*	O2a2b2a2a1*
D1a2a1a1a*	L1a1a2a*	O2a2b2a2b*
D1a2a1a1b*	L1a1a2b*	O2a2b2a2b1*
D1a2a2a1a*	L1a1a1a	O2a2b2*
D1a2a2a1a2a	L1a1b1b*	O2a2b2b1*
D1a2a2a2*	L1a1b2a1*	O2a2b2b1a1*
D1a2a2a2b*	L1a1b2b*	O2b1a*
D1a2a2b2a*	L1a1b3a1*	O2b1a1*
D1a2a2b*	L1a1b3a1a1*	O2b1a1a*
D1a2b2*	L1a1b3a1a1b	O2b1a1a1*
D1a2b1a1a1*	L1a1b3a1a2a	O2b1a1a1a1*
D1a2b1a1a2*	L1a1b3a2a*	O1a1a*
D1a2a1a1a2*	L1a1b3a2b*	O1a1a1a*
D1a2a1a1a*	L1a1b3a*	O1a1a1a1*
D1a2a1a2a*	L1a1b3b*	O1a1a1a1*
D1a2a1a2b*	L1a1b3c1*	O1a1a1a1a*
D1a2a1b1*	L1a2*	O1a1a1a1a1*
D1a2a1b1a*	L1a2a1a*	O1a1a1a1*
D1a2a1*	L1a2a1a1a1*	O1a1a1*
D1a2a1*	L1a2a1a1a2*	O1a1a2*
D1a2a2*	L1a2a1a1a2*	O1a1a2a1*
D1a2a2a*	L1a2a1a1b*	O1a1a2b1*
D1a2a2b*	L1a2a1b*	O1a2a2b1a*
D1a2b*	L1a2a1a1*	O1a1a1a1a1a
D1a2b1*	L1a2a1a2*	O1a1a1a1a1b
D1a2b1a1*	L1a2a1a2*	O1a1a1a1a1c
D1a2b1a1a1*	N*	O1a1a1a1a1d
E1a2b1a1a1a	N1a*	O1b1a1a1a2b
E1a2b1a1a1c	N1a1a1a1a1a	O1a1a1b1*
E1a2b1a1a2*	N1a1a1a1a2a	O1a1a1a1g*
E1a2b1a1a2b*	N1a1a1a1a3a	O1a1a1a1a1a
E1a2b1a1b*	N1a1a1a1a4*	O1a1a1a1a1b
E1a2b1b*	N1a1a1a1a4a	O1a1a1a1a2*
E1b1a1a1a1a	N1a1a2*	O1a1a1a1a1b
E1b1a1a1a1c	N1a2a*	O1a1a1a1a1b
E1b1a1a1a2a	N1a2a1*	O1a1a1a1a3*
E1b1a1a1a2b	N1a2a1a1*	O1a1a1a1a1a
E1b1a1a1a3*	N1a2a1a2*	O1a1a1a1a2*
E1b1a1a1b*	N1a2a1a3*	O1a1a1a1a1a
E1b1a1a1a1*	N1a2b1b1*	O1a1a1a1a1b
E1b1a1a1a1a	N1a2b3*	O1a1a1a1a1*
E1b1a1a1a1b	N1a3*	O1a1a1a1a2*
E1b1a1a1a2*	N1b1*	O1a1a1a1a1*
E1b1a1a1a2*	N1b1a1*	O1a1a1a1a1a
E1b1a1a1b1*	N1b1a1a*	O1a1a1a1a1*
E1b1a1a1b2*	N1b1a1b*	O1a1a1a1a2*
E1b1a1a1c1*	N1b1a2*	O1a1a1a1a1*
E1b1a1a1c2*	N1b1a2a*	O1a1a1a1a*
E1b1a1a1d*	N1b1a2b*	O1a1a1a1a*
E1b1a1a1e*	N1b1a2b1*	O1a1a1a1a*
E1b1a1b*	N1b1a2b1a*	O1a1a2b1b1*
E1b1a1b2*	N1b1b*	O1a1a2b1b2*
E1b1a1b3*	N1b1b*	O1a1a3a*
E1b1a1a1a	N1b2*	O1a1a3a1*
E1b1a1a1a1c	N1b2a*	O1a1a3b*
E1b1a1a1a2a	N1b2a1*	O1a1a3b*
E1b1a1a1a1b	N1b2a1a*	O1a1a3b*
E1b1a1a1a2b	N1b2b*	O1a1b1*
E1b1a1a2*	Nb2*	O1a2a1a1*
E1b1a1a1a1*	O1*	O1a2a1a1a*
E1b1b1a1a1c	O1a*	O1a2a*
E1b1b1a1a1c	O1a1a1a1*	O2a1a*
E1b1b1a1a1g	O1a1a1a1a*	O2a1c1b1*
E1b1b1b1b*	O1a1a1a1a1a	O2c2*
E1b1b1b1b1a	O1a1a1a1a1b	O2b1a1*
E1b1b1b1b1b	O1a1a1a2*	O2b2*
E1b1b1b1b2*	O1a1a1a2a	O2b2a1a*
E1b1b1b2a1a	O1a1a1a3*	O2b3a*
E1b1b1b2b2a	O1a1a1a3a*	R1a1a1a1b1a
E2b1a1c*	O1a1a1b*	R1a1a1b1a1a
E2b1a2*	O1a1a1b1a1*	R1a1a1b1a2b
E2b2*	O1a1a1b1b*	R1a1a1b1a2c
E2b2b*	O1a1a1b1b1a	R1a1a1b1a2a
F*	O1a1a1b1b1b	R1a1a1b1a4*
F2*	O1a1a2a1*	R1a1a1b1a4b
F4*	O1a1a2a1a*	R1a1a1b2*
G1a1*	O1a1a2a1a1*	R1a1a1b2a1*
G1a1a2*	O1a1a2a1a1a1*	R1a1a1b2a1a
G1a1a2a1*	O1a1a2a1a1b	R1a1a1b2a2a
G1a1a2b*	O1a1a2a1a2*	R1a1a1b2b2b
G2a1a2a*	O1a1a2a1a3*	R1a1a1b2b2a*
G2a1a2a21*	O1a1a2a1a3a	R1a1a1b2b2b
G2a2a1a2a1a	O1a1a2a1b1*	R1a1a1b2a4b
G2a2a1a2a1b	O1a1a2a2*	R1a1a1b2a1*
G2a2b1a2*	O1a1a2a2a1*	R1a1a1b2b1*
G2a2b1b1*	O1a1a2a2a1a	R1b*
G2a2b1a1a1a	O1a1a2a*	R1b1a1a1*
G2a2b1a1a1b	O1a2a1*	R1b1a1a1b*
G2a2b1a1a1c	O1a1a2b1*	R1b1a1a1b*
G2a2b1a1a1b	O1a1a2c*	R1b1a1b1a2*
G2a2b1a1a1c	O1a1b*	R1b1a1b1a2a
G2a2b2a1a1d	O1a1b1a1a*	R1b1a1b1b*
G2a2b2a2a*	O1a1b1a2*	R1b1a1b1b3a
G2a2b2a2a2*	O1a2a*	R1b1a2*
G2b1*	O1a2a2*	R1b1a2a1a*
H1a1a1b*	O1a3*	R1b1a2b1*
H1a1a1c*	O1b1*	R1b1b2a2b1*
H1a1a1a*	O1b1a*	R1b2b*
H1a1a4a1*	O1b1a1a1*	R1b2b1*
H1a1a4a2*	O1b1a1a1a1*	R1b2b2*
H1a1a4b*	O1b1a1a1a1a	R2a2b1*
H1a1a4a1a*	O1b1a1a1a1b	R2a2b1b1*
H1a1a4a1b*	O1b1a1a1a2*	R2a2b1b2a1a
H1a1a4b2*	O1b1a1a1a2a	R2a2b1b2b2a
H1a1a4b3a*	O1b1a1a1b*	R2a2b1b2b2b
H1a1a4b2a1*	O1b1a1a1b1*	R2a2b1b2b3*
H1a1a4b2a2*	O1b1a1a1b1a	R2a2b1b2b3b
H1a1a4b2b1b	O1b1a1a1b1b	R2a2b1b2c*
H1a1a4b2b2*	O1b1a1a1b1c	R2a2b1b2c1*
H1a1a4b2c1a	O1b1a1a1b2*	R2b1*
H1a1a4b2c2*	O1b1a1a1*	T1a1a*
H1a1a4b2c3*	O1b1a1a*	T1a1a2b2b
H1a1a4b2d*	O1b1a1a1*	T1a2a1a*
H1a1a4b3a1*	O1b1a1a1a1a	T1a2b1a1*
H1a1a4b3a2*	O1b1a1b1a1b	T1a3b2a1*

Level 13 (911 categories)

[illegible]

Level 15 (1124 categories)

[illegible]